Chocolates Sales

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Business Understanding

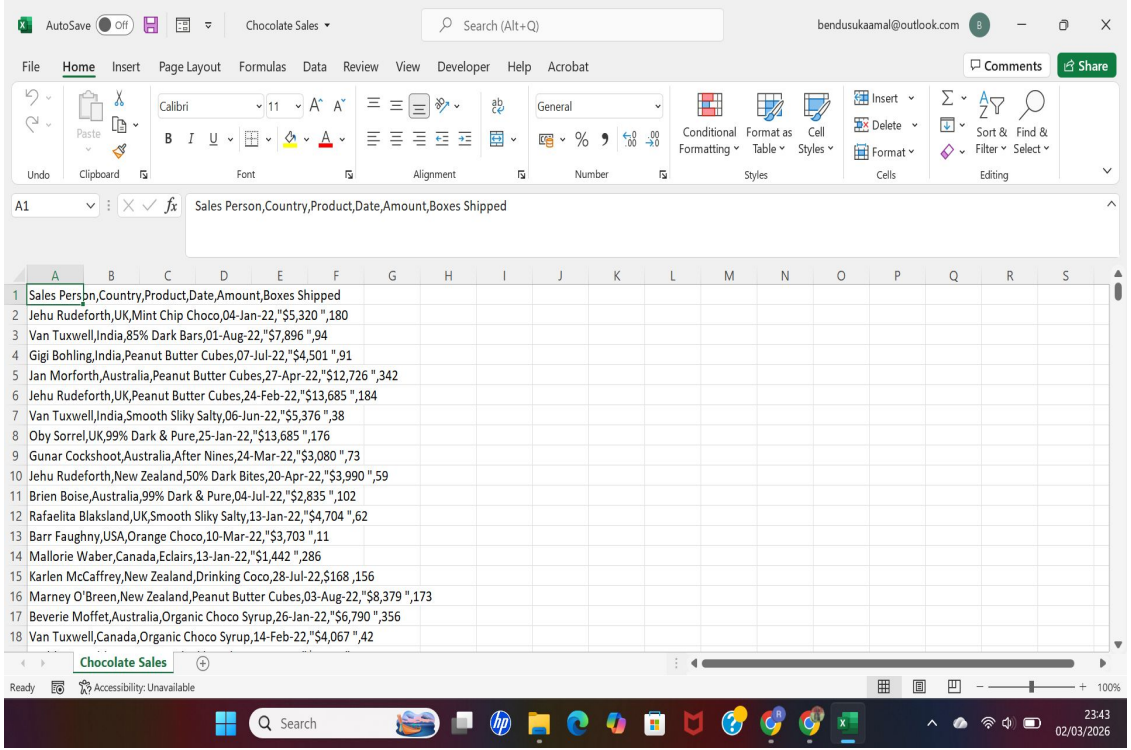
1. Background

   The chocolate industry is so high competitive, with sales performance influenced by product variety, country, and boxes shipped. Understanding sales pattern is crucial moment to improving revenue growth and business optimizing.

2. Problem Statement

   - which country has the highest income?
   - Are there seasonal trends affecting sales performance?
   - How can the company optimize its sales strategy?
   - Which customer has the most purchases?

Dataset



Chocolates Sales have a 6 column (Sales person, Country, Product, Date, Amount, And Boxes Shipped). In this dataset have a 1095 row and when first time to open dataset not clean.

Cleaning Data

```
choco.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1094 entries, 0 to 1093
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Sales Person   1094 non-null   object
 1   Country        1094 non-null   object
 2   Product        1094 non-null   object
 3   Date           1094 non-null   object
 4   Amount         1094 non-null   object
 5   Boxes Shipped  1094 non-null   int64
dtypes: int64(1), object(5)
memory usage: 51.4+ KB
```

This process will check date column still object data type, and after that we will to adjust become datetime type.

In this process i use python to cleaning data and Power BI to transform advanced. First time we must to check with info() to adjust data type in each column

```python
# Ubah ke dalam Float karena kolom masih string
choco['Amount'] = choco['Amount'].replace('[\$,]', '', regex=True)
choco['Amount'] = choco['Amount'].astype(float)

# Ubah ke format datetime
choco['Date'] = pd.to_datetime(choco["Date"])

# Simpan ke dalam variabel baru
choco_clean = choco
choco_clean.head()
```

Process cleaning by remove special character in Amount column and change datatype become to float.

```python
choco_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1094 entries, 0 to 1093
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Sales Person   1094 non-null   object
 1   Country        1094 non-null   object
 2   Product        1094 non-null   object
 3   Date           1094 non-null   datetime64[ns]
 4   Amount         1094 non-null   float64
 5   Boxes Shipped  1094 non-null   int64
dtypes: datetime64[ns](1), float64(1), int64(1), object(3)
memory usage: 51.4+ KB
```

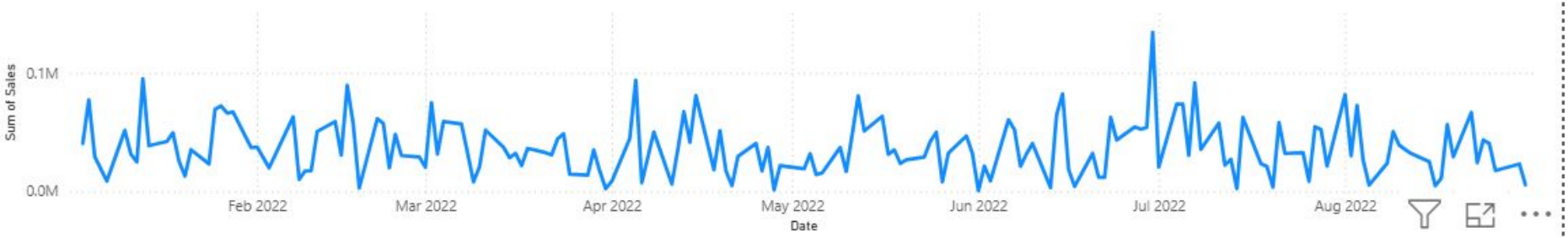
After the process finish, we will to check again with info to ensure type of data adjust by specific column.

Exploratory Data Analysis

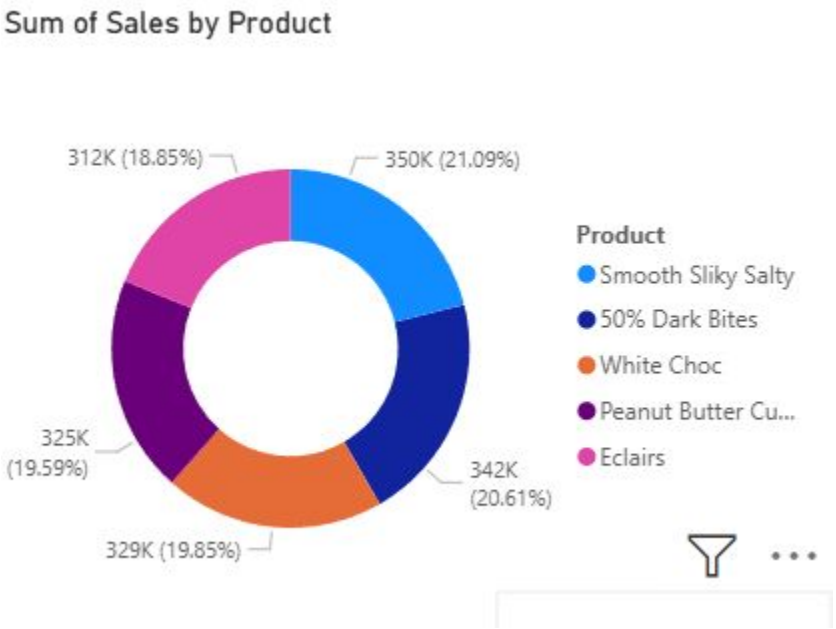
6M	5.65K	177K	161.80
Sum of Sales	**Average of Sales**	Sum of Boxes Shipped	Average of Boxes Shipped

The slide presented a high summary level of exploratory data analysis result. Dataset generate approximately 6M, with an average 5.65K Thousand per transaction.

The company shipped 177 Thousand Box, with an approximately 161.8 per orders

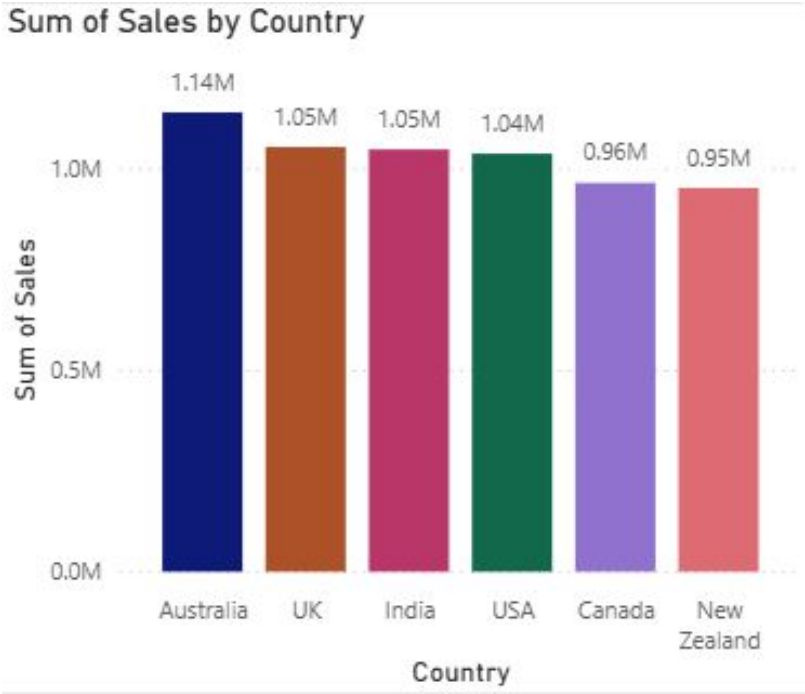


The tren sales has increased in 30 June 2020 by sales 134.063 Thousand. Overall sales remain relatively stable throughout the period. Indicating normal and consistent demand operation performance

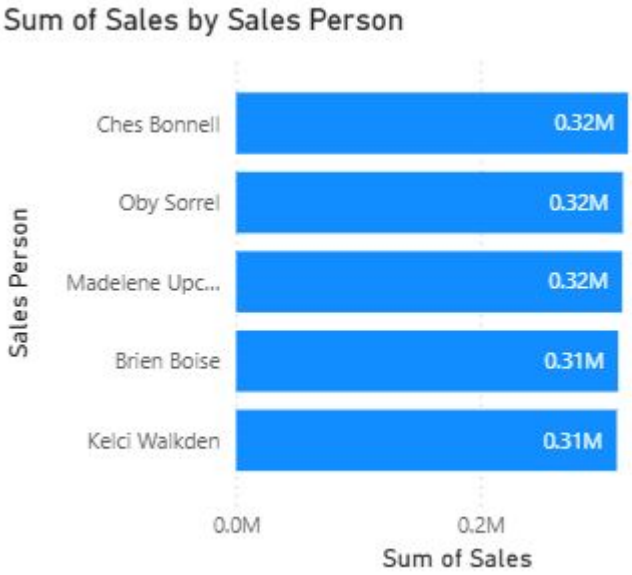


The top 5 best sales product number 1 is a Smooth Sliky Salty with 350 Thousand sales and the lower product is 70% Dark Bites with 211 Thousand sales.

Countries with the most sales is Australia with 1.14 Million and New Zealand is the lowest sales with 0.95 Million



Customer with the most purchase is Ches Borneli with 0.32 Million. Customer indicating balance about purchase product and the range is not so far with other customer.



Insight

- Australia become high sales with the main product is 50% Dark Bites approximately 89 Thousand.
- Ches Borneli become the high sales with 0.32 Million.
- Trend of sales indicating relatively normal and has a increased in 30 June 2022
- Smooth Sliky Salty can be recommended best seller in this company, because become to high sales and beside that be premium product,

Business Recommendation

- The company should prioritize marketing and inventory product top-performing chocolate, as the big contribution total revenue. Beside that optimizing promotion with social media for these product will make boost all sales performance.
- Sales lowest performing countries has be evaluated again to identify potential, local competition, field condition, and distribution coverage. Targeted promotion with social media maybe can increase customer not only old age.